

Technical Analysis Report

US COUNTY EMBEDDINGS & GEOGRAPHIC SIGNAL EVALUATION

Model: BAAI/bge-m3 (1024-dim)

Scope: 3,144 US Counties

Date: July 2026

Executive Summary

Source A's Wikipedia-intro-text county embeddings (**BAAI/bge-m3**, 1024-dim) carry a real but small geographic signal. A Mantel permutation test finds a weak, statistically significant negative correlation between geographic distance and embedding similarity ($r \approx -0.11$, $p = 0.002$). Neither PCA's first component (4.8% of variance) nor K-means clustering ($k = 2$, silhouette ≤ 0.035) finds strong or cleanly interpretable structure beyond that weak geographic effect — **but** the $k = 2$ split is highly reproducible across random seeds ($ARI \approx 0.996$) and its ~11% geographic-coherence deviation from the corpus mean is statistically real (permutation $p \leq 0.001$ both directions), not sampling noise.

A separate prototype found that embedding more article text (additional sections, or an Economy-only section) does not improve differentiation between counties. **Coverage is now complete**: the dataset covers all 3,144 US counties/county-equivalents (the Virginia independent-city gap of 37 cities and a second batch of 19 Census-Gazetteer-vs-Wikipedia-title mismatches have both been backfilled; see §2). Checked against E_macro_extendedProposal.pdf's justification for Source A (§10): the proposal expects intro text to carry distinctive economic-transition narrative; unsupervised analysis instead finds generic formation/demographic boilerplate with only the weak geographic echo above — that specific claim is unsupported, though correlation against real economic variables (Source E/B) is still untested.

Model Choice Rationale: bge-m3 vs. a Frozen LLM Encoder

1. Training Objective Match

bge-m3 is trained with a contrastive/retrieval objective specifically to place semantically similar texts close together under cosine similarity. A frozen decoder-only LLM like Gemma is trained for next-token prediction; its hidden states are not calibrated for similarity search out of the box. Pooling a frozen LLM's hidden states (mean- or last-token-pooling) is a known workaround, not what the model was optimized to produce, and typically needs contrastive fine-tuning (as in dedicated embedding variants) before it is competitive with a purpose-built encoder.

2. Benchmark Performance for This Task Type

Purpose-built embedding models in bge-m3's class lead retrieval/STS benchmarks (e.g., MTEB) at their parameter size; frozen general-purpose LLMs underperform them on semantic-similarity tasks unless separately fine-tuned into an embedding checkpoint — at which point they are no longer "frozen."

3. Context Length Fits the Ingestion Design

bge-m3 natively supports up to 8,192 tokens, which is what ingest_source_a.py and the pre-scoping spec's chunking/mean-pooling fallback (pre_scoping.txt §Source A) are built around. A different backbone with a shorter native context would force chunking on far more articles than currently needed.

4. Compute Footprint

bge-m3 (568M params) runs on CPU at usable throughput for ~3,144 short texts (see README.md's note that MPS/GPU auto-selection was *slower* than CPU for this model on short inputs). A multi-billion-parameter decoder LLM would cost substantially more compute for the same end product — a single fixed vector per county — with no generation capability actually used.

5. Single Deterministic Forward Pass

Producing an embedding is one forward pass with pooling, not autoregressive sampling, which fits this project's reproducibility requirements (fixed seeds, deterministic outputs elsewhere in the pipeline) more naturally than adapting a generation-oriented model into an embedding role.

Granular Empirical Findings & Outlier Analysis

4 Notable Outlier Pairs

The five most similar pairs among all top-quartile-distance pairs ($\geq$ 75th percentile, $\geq$ ~1,940 km apart; full ranked list in `source_a_similarity_pairs.csv`):

Rank	Pair	Similarity	Distance (km)	Why Similar Despite Distance
1	Lincoln County, KS ↔ Lincoln County, OR	0.845	2,207	Shared name <i>and</i> shared namesake (Abraham Lincoln) — both articles use near-identical templated phrasing ("named for/after Abraham Lincoln, 16th president of the United States").
2	Montgomery County, AL ↔ Stutsman County, ND	0.833	1,967	No name in common; both follow the same "county seat is X ... county comprises/included in the X metropolitan/micropolitan area" infobox-style sentence almost verbatim.
3	Stutsman County, ND ↔ Williamsburg County, SC	0.833	2,194	Both are short, minimally-elaborated articles (population + county seat + founding year, no distinguishing detail) — they converge because there is little content to differentiate them, not because of shared subject matter.
4	Franklin County, ME ↔ Franklin County, NE	0.826	2,384	Same mechanism as #1: shared name and shared namesake (Benjamin Franklin) reproduces the same "named for Benjamin Franklin" sentence in both articles.
5	Stutsman County, ND ↔ Providence County, RI	0.826	2,234	Same mechanism as #3 — Stutsman's terse, template-only intro sits close to the corpus's generic-boilerplate centroid, so it reads as "similar" to several otherwise-unrelated counties.

Two distinct surface-level mechanisms produce these matches:

- **(a) Shared eponym** — two counties named after the same historical figure reuse the same "named for X" sentence almost verbatim (#1, #4);
- **(b) Generic-boilerplate convergence** — a short, unelaborated article (Stutsman County, ND appears in 3 of the top 5) sits close to the corpus's templated-boilerplate centroid and reads as similar to any other equally generic article, regardless of subject (#2, #3, #5).

Neither mechanism reflects real economic or geographic kinship — both are consistent with finding #1 (geography matters a little on average; individual pairs can still be highly similar for reasons unrelated to distance) and reinforce the report's broader theme that the corpus is dominated by generic formation/demographic boilerplate rather than distinctive content (§10).

5 Cross-Seed K-Means Stability

Re-running K-means at $k = 2$ across 5 seeds (42, 7, 123, 2024, 99) on the full corpus gives silhouette mean = 0.0343, std = 0.0008, and pairwise Adjusted Rand Index across all 10 seed pairs of mean = 0.09955 (range 0.9916–1.0000). **The $k = 2$ partition is highly stable and reproducible** — it is not an artifact of one random seed.

6 Cluster-Coherence Permutation Test

For the $k = 2$ clustering (999 permutations of county-to-cluster labels, preserving cluster sizes), both clusters' ~11% deviation from the corpus mean distance is more extreme than every one of the 999 random relabelings produced ($p \leq 0.001$, both directions). **The geographic-coherence split is statistically real, not chance** — but this does not upgrade its practical size: silhouette stays ≤ 0.035 , so it is a real, reproducible, but small effect, detectable mainly because of the large sample size, not because the clusters are well-separated. **Do not describe the two clusters as "regions."**